

Math 12 – Exam 2 (Harder Practice)

Instructions: Show all work. Simplify completely and give exact answers.

Simplify the following expressions.

(a)

$$\frac{(2x^{-3}y^4)(6x^5y^{-2})}{3x^2y}$$

(b)

$$\left(\frac{-3x^2y^{-3}}{9x^{-1}y^2}\right)^{-1}$$

Expand and simplify.

$$(2x - 3)^2 - (x + 4)(x - 5)$$

Polynomial Division

Find the quotient and remainder:

$$\frac{12x^3 - 7x^2 + 2x - 9}{3x + 2}$$

Factor completely.

(a)

$$6x^3 - 24x + 3x^2 - 12$$

(b)

$$8x^4 - 72$$

Solve the quadratic equation (do not use the quadratic formula).

$$2x^2 - 11x + 12 = 0$$

Application Problem

A rock is dropped from a height of 196 feet. Its height is given by:

$$h(t) = -16t^2 + 196$$

How long will it take for the rock to hit the ground?

Domain of a Rational Function

Find the domain in interval notation:

$$f(x) = \frac{3x + 1}{x^2 - 5x - 14}$$

Simplify the complex fraction.

$$\frac{1 - \frac{2}{x}}{1 + \frac{3}{x}}$$

Add and simplify.

$$\frac{2}{x(x-4)} + \frac{5}{x-4}$$

Solve the rational equation.

$$\frac{3}{y-2} = \frac{6}{y+4}$$

Word Problem

A fruit drink mixture uses 3 liters of concentrate for every 7 liters of water. If the total mixture is 120 liters, how many liters of water are needed?

Extra Factoring Practice

1.

$$4x^3 + 20x^2 - 3x - 15$$

2.

$$18x^4 - 50$$